

SOUTH HAVEN, MI, USA

WMO: 722343

Lat: 42.351N

Lon: 86.256W

Elev: 666

StdP: 14.35

Time Zone: -5.00 (NAE)

Period: 05-19

WBAN: 54826

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 5.9	(c) 10.2	(d) -2.1	(e) 5.0	(f) 7.3	(g) 2.5	(h) 6.4	(i) 12.3	(j) 23.8	(k) 30.2	(l) 21.0	(m) 29.0	(n) 6.7	(o) 200	(p) 0.492

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 17.7	(c) 88.0	(d) 72.7	(e) 85.2	(f) 71.4	(g) 82.3	(h) 70.2	(i) 75.8	(j) 83.3	(k) 74.3	(l) 81.4	(m) 72.8	(n) 79.6	(o) 8.6	(p) 230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
73.3	126.9	79.8	72.2	122.0	78.9	70.4	114.9	77.1	39.7	83.5	38.2	81.4	36.8	79.6	81.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 20.6	(b) 18.4	(c) 16.6	(e) -3.0	(f) 91.2	(g) 6.6	(h) 2.9	(i) -7.7	(j) 93.3	(k) -11.6	(l) 95.0	(m) -15.3	(n) 96.6	(o) -20.1	(p) 98.7		
(4) 20.6	(5) 18.4	(6) 16.6	(7) -3.7	(8) 78.3	(9) 6.3	(10) 2.0	(11) -8.3	(12) 79.8	(13) -12.0	(14) 81.0	(15) -15.5	(16) 82.1	(17) -20.1	(18) 83.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	49.5	26.9	27.7	36.4	47.2	57.7	66.9	70.7	69.4	63.8	52.5	41.5	32.5		
	DBStd	17.80	9.91	9.73	10.76	9.31	9.21	6.84	6.00	5.29	7.57	8.67	8.66	8.24		
	HDD50	2881	717	626	446	163	29	0	0	0	2	71	282	545		
	HDD65	6238	1181	1045	889	539	265	56	17	20	113	400	706	1007		
	CDD50	2715	1	1	24	78	269	509	641	600	415	148	26	3		
	CDD65	598	0	0	2	4	40	114	194	155	76	13	0	0		
	CDH74	4701	0	0	15	39	395	912	1613	1091	546	90	0	0		
	CDH80	1178	0	0	2	2	101	238	451	230	142	12	0	0		
	WSAvg	6.8	8.4	8.1	7.6	8.1	6.3	5.4	4.9	4.5	5.2	7.1	8.0	8.2		
	PrecAvg	29.2	1.9	1.3	1.9	2.9	2.8	2.8	2.8	3.0	3.3	2.6	2.9	2.2		
(15) Precipitation	PrecMax	46.9	3.5	5.6	4.3	6.1	7.9	7.2	10.6	10.8	12.6	6.3	8.1	5.6		
	PrecMin	20.7	0.1	0.0	0.4	0.6	0.7	0.3	0.6	0.3	0.0	0.4	1.1	0.6		
	PrecStd	5.1	1.0	1.0	0.9	1.2	1.7	1.6	1.8	2.0	2.4	1.4	1.6	1.1		
	PrecStd	5.1	1.0	1.0	0.9	1.2	1.7	1.6	1.8	2.0	2.4	1.4	1.6	1.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	58.0	59.7	75.3	79.0	87.2	90.1	91.3	89.6	88.4	81.5	69.9	60.7		
		MCWB	55.9	53.6	60.0	60.4	70.6	72.5	75.7	72.9	71.6	69.3	57.4	56.4		
	2%	DB	49.9	51.9	66.3	73.2	82.6	86.0	88.1	85.7	84.2	76.6	63.3	53.4		
		MCWB	46.9	47.1	56.6	57.3	66.9	70.6	74.0	72.1	71.2	65.4	54.4	50.7		
	5%	DB	44.0	45.3	59.3	69.7	78.7	82.3	84.6	82.2	80.5	71.8	59.3	48.0		
		MCWB	40.4	41.2	51.8	54.8	64.5	69.6	72.1	71.2	69.0	63.5	52.0	44.4		
	10%	DB	39.3	40.4	53.1	63.5	73.4	79.3	81.9	79.8	76.6	66.9	54.8	43.3		
		MCWB	36.6	36.8	46.5	52.4	62.5	68.0	71.0	70.0	67.5	60.7	49.2	39.7		
	0.4%	WB	55.9	55.1	62.5	64.1	72.2	76.0	78.4	77.4	76.2	71.4	60.9	57.6		
		MCDB	58.7	56.9	73.3	74.4	83.6	85.5	87.4	84.6	83.3	79.9	65.0	59.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	46.9	47.2	58.3	60.0	69.4	73.4	76.2	75.1	73.4	68.1	56.6	50.2		
		MCDB	48.5	51.3	64.5	68.8	79.0	81.8	84.3	81.3	80.2	73.8	62.6	52.9		
	5%	WB	40.9	41.5	53.1	57.1	67.2	71.7	74.7	73.4	71.2	65.1	52.8	44.8		
		MCDB	43.6	45.2	58.7	66.7	75.9	79.6	82.1	79.5	77.4	69.6	58.2	47.0		
	10%	WB	37.3	37.1	46.4	53.5	64.4	69.9	73.0	72.0	69.1	60.7	49.7	40.4		
		MCDB	39.5	40.1	51.8	62.4	71.6	76.6	79.7	77.9	74.4	66.4	55.0	43.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.6	12.3	15.4	18.3	19.4	18.1	17.7	17.6	18.2	16.2	13.0	9.9		
		MCDBR	16.1	18.7	23.7	26.5	24.1	20.6	20.2	19.9	20.9	21.5	18.9	16.8		
		MCWBR	13.9	15.6	16.2	14.7	11.9	9.9	9.8	10.1	9.9	12.8	13.0	14.2		
		MCDBR	15.1	18.2	21.4	23.3	21.7	17.1	17.4	16.1	16.9	18.9	17.1	15.8		
	5% WB	MCWBR	13.6	15.9	15.9	15.2	12.3	9.6	9.8	9.4	9.8	13.7	13.9	14.4		
		MCWBR	13.6	15.9	15.9	15.2	12.3	9.6	9.8	9.4	9.8	13.7	13.9	14.4		
(40) Clear-Sky Solar Irradiance	taub		0.302	0.331	0.345	0.384	0.413	0.428	0.431	0.428	0.400	0.357	0.321	0.305		
	taud		2.416	2.322	2.366	2.286	2.267	2.276	2.269	2.279	2.322	2.434	2.483	2.449		
	Ebn at Noon		267	274	285	280	274	269	267	264	263	263	258	255		
	Edh at Noon		25	32	34	40	42	42	42	40	36	28	23	22		
(44) All-Sky Solar Radiation	RadAvg		400	652	1123	1507	1738	1976	2010	1720	1377	867	512	335		
	RadStd		43	91	95	131	158	129	88	96	118	98	79	46		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	+1.69	N/A	N/A	N/A	N/A
(47) Regional (61 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page